

Mixed Pine Oak Woodlands

PHOTO BY BRAD ALEXANDER, GA DNR



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TRILLIUM PERSISTENS (*TRILLIUM PERSISTENS*)

IMPORTANCE TO FORESTRY OPERATIONS

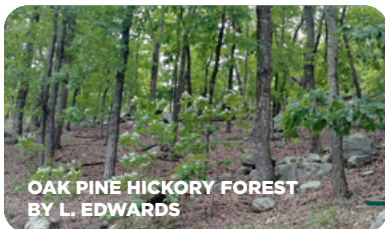
Mixed pine oak forests have long been valued for their hardwoods and merchantable pine, providing diverse timber products that supported local and regional markets. These forests offered operational flexibility as both pine and oak species could be harvested depending on market demands.

Historically, fire played a key role in maintaining the open canopy structure, which favored species diversity and regeneration. However, fire suppression and land-use changes have altered these dynamics, impacting forest composition.

Today, sustainable forestry in mixed pine oak forests relies on prescribed fire and careful harvest planning following BMPs to balance timber production with ecosystem health and wildlife habitat.

HABITAT DESCRIPTION

These communities are characterized by a canopy mixture of upland fire-tolerant oak species (blackjack, post, chestnut, white, black, scarlet, northern red, southern red) and pines (shortleaf, loblolly and, in some communities, longleaf). While historic communities tended toward more open canopies, mixed pine oak woodland communities today may exhibit a range of canopy coverage – from near-complete closure in some forest stands to less than 40% coverage in open canopied woodlands. The mid-story and understory vegetation is limited by canopy closure and fire history, and generally consists of heath shrubs (blueberry, sparkleberry, rhododendron, azalea, mountain laurel), hawthorns, and viburnums. Under more open-canopied conditions, herbaceous groundcover is present and consists of a mixture of sun-loving grasses, sedges, and forbs. Historically, fire was the main driver in these communities; however, other factors included rocky shallow soils, south or west-facing aspects, windthrow, ice storms, and disease.



OAK PINE HICKORY FOREST
BY L. EDWARDS



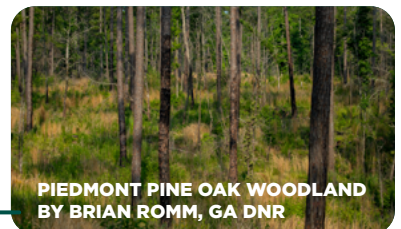
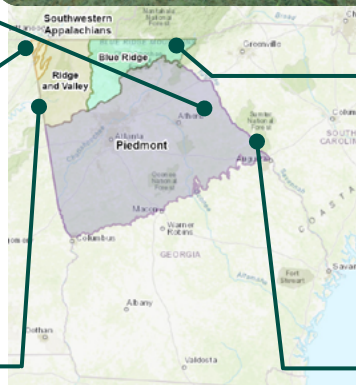
PIEDMONT WOODLANDS, GA
BY ALAN CRESSLER



XERIC PINE WOODLANDS
BY HUGH AND CAROL NOURSE



**MOUNTAIN LONGLEAF PINE
WOODLANDS**
BY ALAN CRESSLER



PIEDMONT PINE OAK WOODLAND
BY BRIAN ROMM, GA DNR

SWAP Priority Habitat	Ecoregion(s)
Mixed Pine Hardwood Forests	Blue Ridge
Montane Longleaf Pine-Hardwood Forest	Piedmont, Southwestern Appalachians/ Ridge and Valley Piedmont
Oak-Pine-Hickory Forest	
Pine-Oak Woodlands and Forest	
Xeric Pine Woodlands	Blue Ridge, Piedmont

THREATS

- **Habitat loss, degradation, and fragmentation, including clearing and conversion of habitat to agriculture, and residential and commercial development**
- **Fire suppression can degrade these habitats by allowing competition from undesirable competing vegetation.**
- **Indiscriminate herbicide use or mechanical vegetation treatments can reduce plant biodiversity and negatively impact ecosystem function.**
- **Overstocked stands can degrade habitat by creating closed canopy conditions and reducing sunlight on the ground.**

CONSERVATION MANAGEMENT RECOMMENDATIONS

- **Use selective spot herbicide and mechanical treatments to control exotic pest plants.**
- **Clean vehicles, tractors, and equipment between sites if invasive species are present. If operations cannot be avoided in areas where invasive species are present, conduct operations in these areas last.**
- **Reduce invasion by woody vegetation utilizing prescribed fire on a 2-3-year rotation or with selective spot applications of herbicide if prescribed fire management is not an option.**
- **Maintain open canopy through selective thinning, if needed.**
- **If implementing prescribed burning, try to use existing boundary roads and natural features as fire breaks to minimize soil disturbance.**

MORE INFORMATION

Visit Georgia Biodiversity Portal for more information about specific species and habitats:

[HTTPS://GEORGIABIODIVERSITY.ORG](https://georgiabiodiversity.org)

Georgia's State Wildlife Action Plan has more information about specific management actions to protect these habitats:

[HTTPS://GEORGIAWILDLIFE.COM/WILDLIFEACTIONPLAN](https://georgiawildlife.com/wildlifeactionplan)

SELECTED PRIORITY SPECIES



See full list of Species
via QR code below

Georgia's Forestry BMP manual is designed to help landowners prevent or minimize impacts to water systems, the manual is available online.

[HTTPS://GATREES.ORG](https://gatrees.org)

View the GA SIC Storyboard
for more information and resources

